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RFO Page Replacement Algorithm

Write a C program to implement the RFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

- The first line of input contains an integer representing the number of pages in the reference string.
- The second line contains the page reference string as a sequence of integers.
- The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

```
Select Language: C (GCC 9.2.0)

1 #include <stdio.h>
2
3 int main() {
4     int n, f;
5
6     printf("Enter number of pages in reference string: \n");
7     scanf("%d", &n);
8
9     int ref[n];
10    printf("Enter the reference string: \n");
11    for(int i = 0; i < n; i++) {
12        scanf("%d", &ref[i]);
13    }
14
15    printf("Enter number of frames: \n");
16    scanf("%d", &f);
17
18    int frames[f];
19    for(int i = 0; i < f; i++) {
20        frames[i] = -1;
21    }
22
23    int index = 0, faults = 0, hits = 0;
24
25    for(int i = 0; i < n; i++) {
26        int found = 0;
27
28        for(int j = 0; j < f; j++) {
29            if(frames[j] == ref[i]) {
30                found = 1;
31                break;
32            }
33        }
34
35        if(found) {
36            hits++;
37        } else {
38            frames[index] = ref[i];
39            index = (index + 1) % f;
40            faults++;
41        }
42    }
43
44    printf("\nTotal Page Faults: %d\n", faults);
45    printf("Total Page Hits: %d\n", hits);
46
47    return 0;
48 }
```

Custom Test Case

▼ Custom Test

Save

Input:

```
7
1 3 0 3 5 6 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Output:

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```